Comprehensive Sensor Architecture, Error Mitigation, and Vehicle Integration for a Formula Student Autonomous Racecar

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***Abstract*—Autonomous and electric Formula Student racecars require a deterministic, low-latency sensor suite capable of operating under severe mechanical vibration, electromagnetic interference (EMI), and dynamic slip conditions. This paper presents an in-depth engineering analysis of the compulsory perception, state-estimation, and actuator-feedback architecture deployed on a Formula Student Driverless (FSD) vehicle. The investigated stack comprises the Hesai Pandar40P 3D mechanical LiDAR, Stereolabs ZED 2i stereo/monocular vision system, VectorNav VN-200 GNSS-Aided Inertial Navigation System (INS), Inertial Measurement Units (IMUs), accelerometer and magnetometer topologies, Hall-effect and optical wheel/ground-speed sensors, Steer-by-Wire (SbW) and Brake-by-Wire (BbW) feedback transducers, and the NVIDIA Jetson AGX Orin central compute unit. Underlying physical principles, error mechanisms, algorithmic compensation, hardware integration, and domestic Indian market sourcing are systematically established.**

***Keywords*—Formula Student Driverless, Hesai Pandar40P, VectorNav VN-200, Stereolabs ZED 2i, NVIDIA Jetson AGX Orin, INS, IMU, Hall-effect, Steer-by-Wire, Brake-by-Wire, sensor fusion.**

I. Introduction

In Formula Student competitions such as Formula Bharat and Formula Student Germany (FSG), racecar performance depends on closing high-frequency control loops across both vehicle dynamics and autonomous navigation [1]. The sensor architecture is divided into two functional domains: *exteroceptive sensors*, which perceive external track boundaries marked by blue, yellow, and orange traffic cones; and *proprioceptive sensors*, which estimate internal vehicle states including three-dimensional attitude, linear velocity, wheel slip ratio, hydraulic brake pressure, and steering rack displacement.

As illustrated in Fig. 1, raw transducer data are ingested by a heterogeneous compute hierarchy. High-bandwidth perception streams from the LiDAR and camera interface directly with the NVIDIA Jetson AGX Orin, whereas safety-critical chassis transducers interface with a real-time microcontroller (Vehicle Control Unit / Autonomous System Brake controller) over deterministic Controller Area Network (CAN) buses.

II. Exteroceptive 3D LiDAR: Hesai Pandar40P

A. Working Principle and Channel Architecture

The Hesai Pandar40P is a 40-channel mechanical spinning Light Detection and Ranging (LiDAR) sensor operating at a Class-1 eye-safe near-infrared wavelength of $\lambda = 905$ nm [2]. It measures three-dimensional radial distance d via direct pulsed laser Time-of-Flight (ToF), measuring the elapsed time Δt between photon emission and avalanche photodiode (APD) detection:

$$d = (c \cdot \Delta t) / 2 \tag{1}$$

where $c \approx 3 \times 10^8$ m/s is the speed of light. Rotating at 10 Hz or 20 Hz across a 360° horizontal field of view (HFOV) with a 0.2°–0.4° horizontal resolution, the unit achieves a range of 200 m (at 10% target reflectivity) and generates 720,000 points/s in single-return mode (1.44 million points/s in dual-return mode). Crucially for Formula Student, its 40 vertical channels (−25° to +15° VFOV) are non-uniformly spaced, concentrating a dense 0.33° vertical angular resolution between −6° and +2° near the horizon where 0.3 m tall track cones reside at distances of 15–50 m.

B. Error Mechanisms and Algorithmic Mitigation

1) Rolling scan motion distortion: Because a 10 Hz mechanical sweep takes 100 ms to complete, vehicle longitudinal motion and yaw rotation during the sweep warp the resulting point cloud. This is solved by point-wise motion deskewing: using 400 Hz linear velocity and angular rate updates from the VectorNav VN-200 INS, each point timestamped at t_i is rigidly transformed into the scan-start coordinate frame t_0.

2) Pitch-induced ground-plane false positives: During aggressive braking (>1.5 g), chassis pitch angles cause forward laser rings to strike the asphalt closer to the nosecone. An IMU-seeded Random Sample Consensus (RANSAC) plane-fitting filter dynamically removes ground points prior to Euclidean cone clustering.

3) Airborne dust and rain returns: Operating the Pandar40P in Dual-Return mode records the strongest and last photon reflections, while a Statistical Outlier Removal (SOR) filter discards isolated mid-air scatter points.

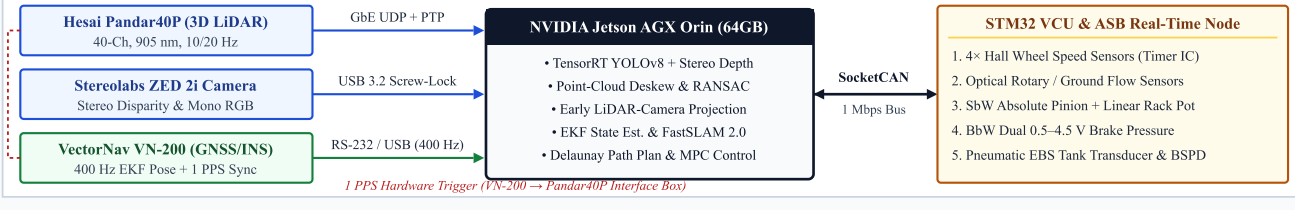


Fig. 1. Complete Formula Manipal Driverless (FM-DV) sensor fusion, IEEE 1588 PTP / 1 PPS synchronization, and dual-CAN bus integration

TABLE I
COMPARATIVE ANALYSIS OF ACCELEROMETER, MAGNETOMETER, AND ODOMETRY TRANSDUCER PHYSICS IN FORMULA STUDENT APPLICATIONS

Transducer Class	Governing Physical Principle	Bandwidth / Range	Primary Racecar Use Case	Dominant Error Mode
MEMS Capacitive Accelerometer	Differential capacitance shift $C = \varepsilon_0\varepsilon_r A/d$ of silicon proof mass	0 Hz (DC) – 400 Hz; ±2 *g* to ±16 *g*	VN-200 INS state estimation, pitch/roll gravity reference, G-G diagram	Thermal bias walk, chassis vibration rectification
Piezoresistive Accelerometer	Strain-induced resistance shift $\Delta R/R = GF\cdot\varepsilon$ in silicon Wheatstone bridge	0 Hz (DC) – 2.5 kHz; ±50 *g* to ±500 *g*	High-energy impact/crash recording and structural load logging	High temperature coefficient of resistance
Piezoelectric (IEPE) Accelerometer	Charge generation $Q = d_{33}\cdot F$ across stressed PZT/quartz crystal	1 Hz – 10 kHz (No static DC response)	Gearbox bearing knock, inverter/motor NVH, suspension chatter	Low-frequency charge leakage, cable triboelectric noise
AMR / TMR Magnetometer	Anisotropic/tunneling magnetoresistance shift with field angle θ	0 – 200 Hz; ±2 to ±8 Gauss	VN-200 static heading initialization and yaw drift bounding	Hard-iron & soft-iron distortion, HV current EMI[a]
Active Hall-Effect Speed Sensor	Lorentz force charge deflection $V_H = IB/(net)$ with Schmitt trigger	0 Hz (zero-speed) – 15 kHz; 0.5–1.5 mm gap	4-corner wheel speed odometry, TCS slip ratio, BLDC commutation	Hub bearing axial runout, wheelspin/lockup slip
Optical Rotary / Flow Sensor	Photodiode pulse interruption (A/B/Z) or spatial surface correlation	0 – 100 kHz (encoder); 250 Hz (ground flow)	Steer-by-Wire motor feedback; slip-free vehicle velocity (v_x, v_y)[b]	Lens occlusion by mud, brake dust, water, or grease

[a]High-voltage tractive system cables carrying up to 300 A DC generate localized magnetic fields exceeding Earth's ambient ~0.5 Gauss field.
[b]Optical flow ground sensors provide direct vehicle slip angle $\beta = \arctan(v_y / v_x)$ independent of tire slip.

C. Vehicle Integration and Indian Market Sourcing

On the Formula Manipal racecar, the Pandar40P is mounted on a CNC-machined aluminum bracket atop the Main Roll Hoop (or upper nosecone structure) to provide an unobstructed forward view while keeping the optical window above tire debris. Power is supplied via an isolated 12 V / 24 V DC-DC converter (consuming 18 W steady-state). The LiDAR connects via an M12 aviation harness to its interface box, streaming UDP packets over Gigabit Ethernet to the NVIDIA Jetson AGX Orin (`HesaiLidar_General_ROS` driver publishing `sensor_msgs/msg/PointCloud2`). Sub-microsecond timestamp alignment is enforced by feeding a 1 PPS (Pulse Per Second) CMOS hardware trigger from the VectorNav VN-200 into the LiDAR interface box alongside IEEE 1588 Precision Time Protocol (PTP). In India, the Hesai Pandar40P is sourced through authorized autonomous distributors such as Corebolt LLP (Hyderabad) on IndiaMART or via Hesai academic sponsorship at ₹3,50,000–₹5,50,000 (or ₹55,000–₹90,000 for refurbished autonomous shuttle units).

III. INERTIAL NAVIGATION (INS), VECTORNAV VN-200, AND IMU PHYSICS

A. Distinction Between IMU and INS (VectorNav VN-200)

A critical systems distinction exists between an *Inertial Measurement Unit (IMU)* and an *Inertial Navigation System (INS)*. An IMU is purely a raw transducer triad comprising a 3-axis accelerometer and a 3-axis gyroscope (often augmented by a 3-axis magnetometer to form a 9-DoF module) that outputs body-frame specific force $\mathbf{f}_b = [a_x, a_y, a_z]^T$ and angular velocity $\boldsymbol{\omega}_b = [\omega_x, \omega_y, \omega_z]^T$. Because dead-reckoning requires double integration of linear acceleration, any uncompensated accelerometer bias b_a or gyroscope bias b_g induces quadratic and cubic position drift over time t:

individual factory thermal polynomial calibration across −40°C to +85°C. Second, high-frequency structural vibrations from the drivetrain are suppressed using internal Coning and Sculling integrals alongside a configurable finite impulse response (FIR) low-pass filter. Third, if the VN-200 cannot be mounted exactly at the vehicle Center of Gravity (CG) and is displaced by lever arm $\mathbf{r}$, centripetal acceleration $\boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r})$ corrupts linear acceleration data; this offset is subtracted inside the VN-200 EKF configuration registers.

2) Vehicle integration and sourcing: The VN-200 Rugged is mounted near the CG on elastomeric vibration-damping grommets, connected to the Jetson AGX Orin over RS-232/USB (`vectornav` ROS 2 package), with its active GNSS antenna mounted on the top of the front roll hoop via a shielded MMCX coaxial cable. It is sourced in India via DigiKey India, Mouser India, or CoreEL Technologies (Bengaluru) for ₹1,45,000–₹1,85,000 (while standalone IMUs like the Bosch BNO085 or Xsens MTi-3 cost ₹2,200–₹48,000 on Robu.in).

IV. TAXONOMY OF ACCELEROMETERS AND MAGNETOMETERS

A. Types of Accelerometers

As summarized in Table I, three primary accelerometer physics are utilized in motorsport engineering:

a) MEMS capacitive accelerometers: Etched silicon interdigitated comb fingers form differential capacitors with fixed plates. Under vehicle acceleration, proof-mass deflection alters plate gap d, changing capacitance. Because they respond down to 0 Hz (static gravity), they are standard inside the VN-200 and ZED 2i for vehicle state estimation.

b) Piezoresistive accelerometers: Silicon cantilever beams implanted with p-type piezoresistors experience resistance changes proportional to mechanical strain. Capable of surviving >500 *g* shocks, they are deployed in motorsport crash data recorders.

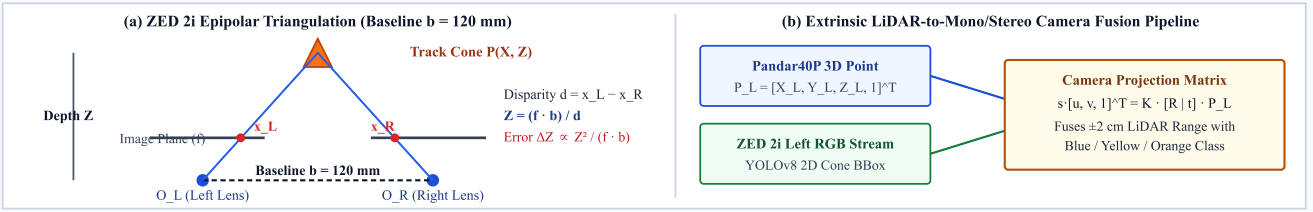


Fig. 2. (a) Epipolar stereo triangulation geometry of the Stereolabs ZED 2i showing baseline b = 120 mm and quadratic error scaling; (b) Rigid extrinsic transformation projecting 3D Hesai Pandar40P cone clusters onto 2D monocular camera bounding boxes.

V. Vision Perception: Stereolabs ZED 2i (Stereo and Monocular Modes)

A. Stereo Camera Working Principle and Depth Triangulation

While 3D LiDAR provides geometric coordinates, it lacks RGB color channels required to distinguish blue (left boundary), yellow (right boundary), and orange (start/finish) Formula Student cones. The **Stereolabs ZED 2i** addresses this by housing dual 4-megapixel 1/3-inch CMOS sensors separated by an optically calibrated baseline of b = 120 mm inside an IP66 dust- and water-resistant aluminum enclosure [4]. Using the wide-angle 2.1 mm focal-length lens (110°(H) × 70°(V) FOV), both sensors capture synchronized frames at up to 1080p @ 30 FPS or 720p @ 60 FPS.

As depicted in Fig. 2(a), after lens distortion rectification aligns both images onto coplanar epipolar scanlines, a 3D cone point $P(X, Y, Z)$ projects onto horizontal pixel coordinates x_L and x_R. Using Stereolabs' GPU-accelerated Neural Depth engine to compute pixel disparity $d = x_L - x_R$, metric depth Z and theoretical depth uncertainty ΔZ are governed by:

$$Z = (f \cdot b) / d\,, \quad \Delta Z = (Z^2 / (f \cdot b)) \cdot \Delta d \tag{4}$$

where f is focal length in pixels and Δd is sub-pixel matching error. Additionally, the ZED 2i integrates an internal 400 Hz 9-DoF IMU, barometer, magnetometer, and thermal sensors to execute Visual-Inertial Odometry (VIO).

B. Monocular Camera Mode and Sensor Fusion Use Case

A monocular camera consists of a single optical sensor that maps 3D world coordinates onto a 2D pixel plane $[u, v, 1]^T$ via the camera intrinsic matrix **K**. In the Formula Manipal pipeline, the ZED 2i operates concurrently as a *calibrated monocular camera* by extracting its rectified left RGB image stream (`zed_node/left/image_rect_color`). Because a single monocular image loses depth scale along the optical ray, distance estimation in standalone mono mode requires assuming a flat track surface via a planar homography transformation (Inverse Perspective Mapping), which breaks down when the car pitches or encounters track banking.

Instead, as shown in Fig. 2(b), we combine the monocular stream with 3D LiDAR point clouds$\mathbf{P}_L$ using rigid extrinsic calibration parameters (rotation matrix **R** and translation vector **t**):

$$s \cdot [u, v, 1]^T = \mathbf{K} \cdot [\mathbf{R} \mid \mathbf{t}] \cdot [X_L, Y_L, Z_L, 1]^T \tag{5}$$

where s is the projective scale factor. TensorRT-accelerated YOLOv8 inference runs on the monocular frame to classify cone colors, while projected Pandar40P clusters supply close-range cones (0.5–12 m, where vertical LiDAR beams may overshoot short cones) and transitions seamlessly to LiDAR-projected depth for 12–40 m cones.

2) Specular glare and rolling-shutter distortion: Harsh Indian mid-day sunlight creates specular reflections on hot tarmac that wash out yellow cones. We utilize the **ZED 2i Polarizer variant** (integrated linear polarizing filter) and configure dynamic Auto-Exposure Region of Interest (ROI) restricted to the lower 60% of the frame (excluding bright sky). High-frequency vibration blur is minimized by mounting the camera on the Main Roll Hoop with Sorbothane damping pads and enforcing short exposure times (<2 ms).

3) USB 3.2 bandwidth stability and sourcing: Standard USB-C connectors vibrate loose on a racecar; we mandate the industrial **USB 3.0 Type-C Dual Screw-Lock cable** bolted directly into the ZED 2i housing and routed to the Jetson AGX Orin. In India, the ZED 2i is sourced directly from Robu.in, ThinkRobotics, or Zbotic for ₹96,599 (standard) to ₹1,03,549 (with polarizer, 2.1 mm lens, inclusive of GST).

VI. Central Compute Engine: NVIDIA Jetson AGX Orin

A. Heterogeneous Compute Architecture

The **NVIDIA Jetson AGX Orin (64GB Developer Kit)** serves as the central autonomous processing unit [5]. It delivers up to **275 TOPS** (INT8) of AI compute via a 2048-core NVIDIA Ampere architecture GPU with 64 Tensor Cores, a 12-core Arm Cortex-A78AE v8.2 64-bit CPU, two NVIDIA Deep Learning Accelerators (NVDLA v2.0), a Vision Accelerator (PVA v2.0), and 64 GB of 256-bit LPDDR5 memory operating at 204.8 GB/s bandwidth. Running Ubuntu 22.04 LTS with a PREEMPT_RT real-time Linux kernel patch and ROS 2 Humble, it partitions workloads across dedicated silicon: YOLO cone detection runs in FP16/INT8 on NVDLA cores, ZED 2i CUDA stereo matching and point-cloud deskewing run on the Ampere GPU, and Graph-SLAM (`iSAM2` / `FastSLAM 2.0`), Delaunay path planning, and Model Predictive Control (MPC) execute on isolated CPU cores via `taskset` CPU affinity.

B. Error Mitigation, Car Integration, and Sourcing

1) Thermal throttling and voltage brownouts: At full 60 W TDP inside a sealed IP65 electronics enclosure in 40°C Indian ambient conditions, thermal throttling degrades frame rates. We mitigate this by bolting the Orin heatsink to an external finned aluminum cold-plate with ducted ram-air cooling and locking clock frequencies via `sudo nvpmodel -m 0` and `sudo jetson_clocks`. To prevent reboots when high-current brake or steering actuators fire on the Low-Voltage (LV) bus, the Orin is powered through an isolated 9–36 V to 19 V automotive DC-

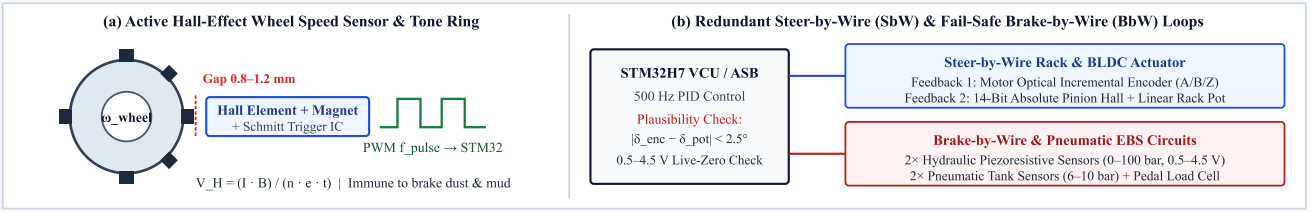


Fig. 3. (a) Active Hall-effect wheel speed probe reading a ferromagnetic hub tone ring; (b) Dual-redundant Steer-by-Wire (SbW) and fail-safe Brake-by-Wire (BbW) / Emergency Brake System (EBS) sensor feedback architecture.

VII. Wheel Speed, Hall-Effect, and Optical Sensors

A. Hall-Effect Physics and Wheel Speed Odometry

When a semiconductor wafer carrying bias current I is exposed to a perpendicular magnetic flux density B, the Lorentz force $\mathbf{F} = q(\mathbf{v} \times \mathbf{B})$ deflects charge carriers toward one edge of the crystal, producing a transverse Hall voltage V_H [6]:

$$V_H = (I \cdot B) / (n \cdot e \cdot t), \quad v_{wheel} = (f_{pulse} / N_{teeth}) \cdot 2\pi r_{eff} \quad (6)$$

where n is carrier density, e is elementary charge, and t is wafer thickness. In our **Wheel Speed Sensors (WSS)**, an active biased Hall probe (containing an internal permanent back-biasing magnet and Schmitt trigger IC) is mounted radially facing a laser-cut 24-tooth steel tone ring bolted to each of the four wheel hubs, as shown in Fig. 3(a). Unlike passive inductive (variable reluctance) sensors whose AC amplitude drops to zero at low speeds, active Hall sensors output a crisp 0–5 V digital square wave of frequency f_{pulse} down to true zero speed. Beyond wheel hubs, non-contact ratiometric Hall sensors are also deployed in the Accelerator Pedal Position Sensor (APPS) and inside our Brushless DC (BLDC) drive motors for rotor commutation.

B. Optical Sensors vs. Hall Effect in Racecar Environments

Optical sensing in a Formula Student car encompasses two distinct hardware devices: (1) **Optical Rotary Encoders**, where a collimated infrared LED shines through a precision etched glass/metal code disk onto quadrature photodiode arrays (channels A, B with 90° phase shift, plus index pulse Z), achieving 1,000–4,096 PPR for high-resolution motor shaft feedback; and (2) **Optical Flow Ground-Speed Sensors** (e.g., Kistler Correvit SF-II or PMW3901 optical array), which image road surface micro-texture to measure true slip-free longitudinal and lateral velocities (v_x, v_y) and vehicle body slip angle $\beta = \arctan(v_y / v_x)$.

Engineering Selection at the Wheel Upright: An unsealed optical disk must never be used at the wheel hub because brake pad carbon dust, mud, tire rubber, and water rapidly occlude the optical path. Conversely, **Hall-Effect sensors are completely immune to non-ferrous dirt, grease, and water**, making them mandatory for wheel speed measurement.

C. WSS Error Mitigation, Integration, and Sourcing

1) Wheelspin, lockup, and bearing runout: During aggressive launch or braking, wheel circumferential speed decouples from true ground speed. We compute the longitudinal slip ratio κ for each corner:

$$\kappa = (\omega \cdot r_{eff} - v_x) / \max(v_x, \omega \cdot r_{eff}) \quad (7)$$

When $|\kappa|$ exceeds a 0.12 threshold, a Mahalanobis-distance gate inside the EKF rejects that wheel's odometry update to prevent state estimation corruption. To prevent hub bearing play under 2 g cornering loads from altering the 0.8–1.2 mm sensor air gap, the Hall probe is mounted *radially* rather than axially. All four sensors wire via shielded twisted-pair cables into STM32 timer Input Capture pins and broadcast over CAN at 100 Hz. Automotive Hall speed sensors (Honeywell SNDH-T4L / Omron E2E / Bosch ABS) cost ₹600–₹3,200 on Robu.in, DigiKey India, and Boodmo, while Omron E6B2-CWZ6C optical rotary encoders cost ₹1,850–₹4,200 on Robu.in.

VIII. Steer-by-Wire (SbW) and Brake-by-Wire (BbW) Sensing

A. Redundant Steer-by-Wire (SbW) Sensor Suite

To execute steering angle commands δ_{cmd} from the Jetson AGX Orin, a BLDC actuator drives the steering rack. Relying solely on the motor's internal incremental optical encoder causes two severe errors: loss of absolute zero-degree reference upon power-up, and unmeasured mechanical backlash in the planetary gearbox. As illustrated in Fig. 3(b), we solve this by placing a **14-bit contactless magnetic Absolute Rotary Encoder** (AMS AS5048A / Bourns EMS22A over SPI) directly on the steering pinion shaft, paired with a redundant **Linear Potentiometer** (Gefran/Miran conductive plastic stroke sensor, 0–5 V ADC) mounted parallel to the steering tie-rod. If the discrepancy $|\delta_{enc} - \delta_{pot}|$ exceeds 2.5° for >50 ms, the VCU flags a plausibility fault.

B. Brake-by-Wire (BbW), EBS, and Live-Zero Fault Detection

Per FSG Driverless rules, the braking system combines an active electric Brake-by-Wire actuator with a passive pneumatic **Emergency Brake System (EBS)** capable of achieving >4.5 m/s^2 deceleration if power is lost [1]. Two **Piezoresistive Hydraulic Pressure Transducers** (0–100 bar, e.g., Bosch PST-F 1 or Wika/Sensata transmitters) monitor the independent front and rear hydraulic brake lines, while two pneumatic pressure transducers (0–10 bar) verify compressed air reservoir pressure during the autonomous pre-drive check, supplemented by a strain-gauge load cell on the pedal box.

1) Solving open-circuit wire faults: A standard 0–5 V sensor reads 0 V when at 0 bar—identical to a severed signal wire! We therefore mandate **0.5–4.5 V ratiometric** (or 4–20 mA current-loop) transducers: atmospheric 0 bar corresponds to 0.50 V, whereas a broken wire or ground short drops the ADC voltage below 0.25 V, allowing the STM32 ASB controller and hardware Brake System Plausibility Device (BSPD) to detect sensor failure within 5 ms and trigger the fail-safe EBS.

TABLE II
MASTER FORMULA MANIPAL SENSOR & COMPUTE BILL OF MATERIALS (BOM), INTERFACES, AND INDIAN MARKET SOURCING

Hardware / Subsystem	Selected Model	Car Interface & Rate	Primary Indian Suppliers	Unit Cost (INR)[a]	Qty / Subtotal
3D Mechanical LiDAR	Hesai Pandar40P (40-Ch)	GbE UDP + PTP/1PPS (20 Hz)	Corebolt LLP (IndiaMART), Hesai Acad.	₹3,85,000[b]	1 × ₹3,85,000
GNSS-Aided INS	VectorNav VN-200 Rugged	RS-232 / USB + 1PPS (400 Hz)	DigiKey India, Mouser, CoreEL Tech	₹1,62,000	1 × ₹1,62,000
Standalone 9-DoF IMU	Bosch BNO085 / Xsens MTi-3	SPI / I2C / CAN (400 Hz)	Robu.in, Evelta, Mouser India	₹2,450 – ₹45k	1 × ₹2,450
Stereo / Mono Camera	Stereolabs ZED 2i (Polarizer)	USB 3.2 Screw-Lock (60 FPS)	Robu.in, ThinkRobotics, Zbotic	₹1,03,549	1 × ₹1,03,549
Central AI Compute	NVIDIA Jetson AGX Orin 64GB	GbE, USB 3.2, Dual SocketCAN	Robu.in, Rptech India, Silverline	₹2,15,000	1 × ₹2,15,000
Wheel Speed (Hall)	Honeywell SNDH-T4L / Omron	STM32 Timer IC → CAN (100 Hz)	DigiKey India, Robu.in, Boodmo	₹1,850	4 × ₹7,400
Optical Rotary Encoder	Omron E6B2-CWZ6C (2000 PPR)	Quadrature A/B/Z Timer (1 kHz)	Robu.in, RS Components India, IndiaMART	₹2,600	1 × ₹2,600
Optical Flow Sensor	PMW3901MB / Correvit SF-II	SPI / CAN Ground Vel (250 Hz)	Robu.in (PMW3901) / Kistler India	₹1,650	1 × ₹1,650
SbW Angle & Rack Pos.	Bourns EMS22A + Miran KTR Pot	SPI + 0–5 V ADC → CAN (500 Hz)	Mouser India, Robu.in, IndiaMART	₹6,800 (Set)	1 × ₹6,800
BbW Hydraulic Pressure	Bosch PST-F 1 / Wika 0–100 bar	0.5–4.5 V Ratiometric ADC (500 Hz)	RS Components India, Element14, Mouser	₹4,500	2 × ₹9,000
EBS Pneumatic & Load	Baumer 0–10 bar + 100 kg Load Cell	4–20 mA / HX711 → ASB Node	IndiaMART, Robu.in, Element14 India	₹3,800 (Set)	1 × ₹3,800

[a]Prices reflect verified October 2026 Indian retail/distributor listings inclusive of 18% GST where applicable. [b]Refurbished autonomous shuttle Pandar40P units are available at ₹65,000–₹90,000, reducing total sensor suite outlay from ₹8,99,249 to ~₹5,80,000.

IX. System-Level Sensor Fusion and Calibration Synthesis

Achieving reliable autonomous lap completion requires unifying all eleven subsystems listed in Table II under a common spatial and temporal framework. Spatially, every sensor frame is referenced to the vehicle rear-axle centerline via a Unified Robot Description Format (`URDF`) tree in ROS 2. Extrinsic 6-DoF calibration between the Hesai Pandar40P and Stereolabs ZED 2i is performed offline using a checkerboard target with circular cutouts, minimizing 3D-to-2D reprojection error via Levenberg-Marquardt nonlinear optimization.

Temporally, the VectorNav VN-200 acts as the master GPS-disciplined clock source, driving a 1 PPS hardware interrupt into the Pandar40P while `linuxptp` (`ptp4l` and `phc2sys`) synchronizes the Jetson AGX Orin system clock over Gigabit Ethernet to within ±10 µs. State estimation fuses 400 Hz VN-200 inertial updates, 100 Hz Mahalanobis-gated 4-wheel Hall odometry, and optical flow velocity inside an Unscented Kalman Filter (`robot_localization`), feeding a drift-free 15-state vehicle pose vector to the Graph-SLAM and Model Predictive Control pipelines.

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